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Python Operators
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=> An operator is a symbol or keyword used to perform operations on values or variables.

=> Operators are special symbols used to perform operations like addition, comparison, assignment, logical checking, and more.

Why do we need operators

** Calculate students marks

** Calculate Total product price (Actual Price + GST)

** Check Login Credentials

** Check Citizen voting eligibility

** Apply Discounts on products based on Coupons

** Credit or Debit Account balance amount

Types of Operators

1) Arithmetic Operators (+, -, *, / , //, %, +=, -=)

2) Assignment Operators (=)

3) Comparison Operators (==, !=, >, >=, <, <=)

4) Logical Operators (and, or, not)

5) Membership Operators (in, not in)

6) Identity Operators (is, is not)

*** Arithmetic operators are used to perform mathematical operations.

```
a = 10
```

```
b = 3
```

```
print(a + b)    # Addition: 13
print(a - b)    # Subtraction: 7
print(a * b)    # Multiplication: 30
print(a / b)    # Division: 3.3333
print(a // b)   # Floor Division: 3
print(a % b)    # Modulus / Remainder: 1
print(a ** b)   # Power: 1000
```

*** Assignment operators are used to assign values to variables.

```
x = 10
```

```
print(x)    # 10
```

```
x += 5      # x = x + 5
```

```
print(x)    # 15
```

```
x -= 3      # x = x - 3
print(x)    # 12
```

```
x *= 2      # x = x * 2
print(x)    # 24
```

```
x /= 4      # x = x / 4
print(x)    # 6.0
```

```
x //= 2     # x = x // 2
print(x)    # 3.0
```

```
x %= 2      # x = x % 2
print(x)    # 1.0
```

```
# additional assignment
balance = 1000
deposit_amount = 500
#balance = balance + deposit_amount
balance += deposit_amount
print("Final balance : ", balance)
```

```
withdraw_amount = 200
#balance = balance - withdraw_amount
balance -= withdraw_amount
print("Final balance : ", balance)
```

*** Comparison operators are used to compare two values. They return True or False.

```
a = 10
b = 20
```

```
print(a == b)    # Equal to: False
print(a != b)    # Not equal to: True
print(a > b)     # Greater than: False
print(a < b)     # Less than: True
print(a >= b)    # Greater than or equal to: False
print(a <= b)    # Less than or equal to: True
```

*** Logical operators are used to combine multiple conditions.

```
age = 25
salary = 50000
```

```
print(age > 18 and salary > 30000)
# True because both conditions are True
```

```
print(age > 18 or salary > 80000)
# True because one condition is True
```

```
print(not age > 18)
# False because age > 18 is True, not True becomes False
```

*** Membership operators are used to check whether a value is available inside a sequence like list, tuple, string, etc.

```
students = ["Ashok", "Ramesh", "Suresh"]
```

```
print("Ashok" in students)    # True
print("Kiran" in students)    # False
```

```
print("Kiran" not in students)    # True
```

```
*** Identity operators are used to compare memory location of objects.
```

```
a = 10
```

```
b = 10
```

```
print(a is b)    # True
```

```
print(a is not b) # False
```